Auditing Transfer Before Predicting Dexterous Grasp Failure

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Abstract: Failure prediction from simulated grasps depends on proposal transfer, execution and the meaning of the outcome label. We study a HUG-only pipeline with static mapping audits, a fixed Shadow Hand lift-and-hold task and joint-state/command logistic predictors. A 1,040-attempt development collection yields failure AUROC 0.8523 versus 0.6419 for commands alone on a held-out observation; frozen evaluation on 2,080 fresh attempts yields 0.9173 on familiar geometries and 0.5588 on two new geometries pooled together. All 520 original mugs fail during approach before the prediction cutoff, while separate corrected-controller checks trigger warnings on all 20 successful mugs. New retrospective object-held-out fits also show that ranking and warning reliability differ: cube AUROC is 0.8729, but 471/648 successful cube attempts trigger alerts. Static audits accept realization of 32 saved Shadow commands while rejecting strict native-surface preservation; none of these results establishes physical transfer or future-slip warning.

Keywords: Dexterous grasping, failure prediction, simulation validation

1 Introduction and scope

A simulator can generate many failed grasps without generating the failure distribution a deployed predictor needs. Failure may originate in the proposal, human-to-robot retargeting, scene construction, approach controller, or task definition. Treating these outcomes as interchangeable slip events hides the source of a transfer failure. We investigate this problem using Human Universal Grasping (HUG), which predicts a MANO hand pose and wrist placement from RGB-D and supports robot retargeting [1]. Our experiment evaluates the resulting *HUG-to-Shadow execution pipeline*, not HUG's official benchmark or the native human grasp in isolation.

We contribute a layered mapping audit, versioned collection and labeling, and grouped prediction diagnostics exposing distribution shift and temporal target mismatch. Original development/fresh batches retain their frozen protocol; corrected-controller diagnostics remain separate. Static mapping audits add no outcome data.

Definitions and related work. A *proposal* is the generated native pose; an *execution* includes retargeting, embodiment, controller and scene. A *task failure* violates the lift-and-hold rule; failed acquisition is not automatically slip. Calandra et al. [2] study vision/touch outcome prediction; our inputs are simulated joint states and commands. AnyTeleop [3] informs our retargeting software. MuJoCo [4] supplies dynamics without certifying hardware fidelity. Domain randomization [5] motivates varying conditions; joint-bias repetition alone does not test their transfer. We introduce no new forecasting architecture.

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2 From native proposal to declared execution

Audit contract. We separate three questions. First, does the saved native export reconstruct its own hand geometry and frame transforms? Second, does the compiled simulator realize the declared robot command, source joint limits, wrist frame and object scene? Third, does substituting a different anatomy preserve near-object surfaces and their directions? Native reconstruction and independent forward kinematics test numerical implementation; fingertip distances summarize retargeting distortion; surface-distance and proximity checks test relationships that a sparse landmark average can miss. A static pass cannot certify servo tracking, contact forces, lift success or predictor quality. Invalid inputs and runtime errors remain separate from scored task outcomes.

Saved-cohort findings. All 32 saved native exports reconstruct exactly in the recorded 3D audit. A later current-scene bridge accepts static realization of all 32 saved nominal Shadow commands, including source limits and scene checks, with zero physics integration. This accepts the declared robot-input pipeline, not anatomical equivalence. Across the same 160 fingertips, the saved Shadow commands have mean error 6.428 mm and maximum 17.518 mm relative to native fingertip vertices. Detailed surface studies on six diagnostic grasps use 1/2/5 mm proximity bands and direction checks, rejecting the project's stricter native-preservation claim: small mean fingertip error is insufficient to infer preserved hand–object relationships.

A separate native MANO static proxy lowers measured distortion but has contacts and gravity disabled, placeholder inertias and no controller. It is not a validated physical hand or an outcome-collection replacement.

3 Collection, labels and causal inputs

Frozen simulation. The `fixed_grasp_lift_v2` protocol uses a right Shadow Hand with force-limited position servos in MuJoCo. It performs pregrasp, approach, closing, settling, lifting, holding and scoring without outcome-dependent replanning. The commanded lift is 0.15 m; physics steps are 2 ms and control recordings 40 ms. Calibrated 224×224 RGB-D observations, masks and camera transforms feed the released HUG checkpoint without weight training. Content-bound observation and proposal records retain geometry, pose, calibration and source provenance. Capture checks settling before inference.

The development set uses a 0.18 kg cube and 0.2304 kg apple, two observations per geometry and four HUG seeds per observation: sixteen proposals. For each proposal, one zero-bias run and 32 runs at each nonzero amplitude yield $16(1 + 32 + 32) = 1{,}040$ attempts. A direction $\boldsymbol{\xi} \sim U([-1, 1]^{22})$ sets a finger-target bias $a\boldsymbol{\xi}$, fixed within an episode, with $a \in \{0, 0.15, 0.6\}$ radians; wrist targets are unperturbed. Nonzero amplitudes share directions for paired repetitions. Commands are clipped to joint limits, and requested offsets, applied offsets and measured motion are stored separately. This is command bias, not per-frame sensor noise or validated hardware error.

Task labels and event annotations. Success requires qualifying hand–object contact, peak object-center lift of at least 0.14 m, at least 2 s of consecutive sampled HOLD time above 0.12 m, and peak-to-final height loss no greater than 0.02 m. Heights are relative to the initial center. Approach table collision and timeout invalidate task success. The first applicable failure is timeout, approach collision, missing qualifying contact, failed acquisition, drop, then insufficient hold. Invalid input/runtime errors receive null labels rather than task-failure labels. Separate sampled acquisition and height-loss annotations do not overwrite outcomes or assert exact slip onset. Joint states are interval endpoints; contact diagnostics normally precede the final integration substep by 2 ms.

The collection contains 444 successes and 596 failures. Success/failure counts by requested bound are 8/8 at zero, 294/218 at 0.15 rad and 142/370 at 0.6 rad. No collection errors occurred.

Table 1: Failure prediction. Development test: one apple observation, 202 failures/58 successes. Fresh evaluation: all 2,080 additional attempts, no retraining. Lower log loss and Brier score are better.

Evaluation	Model	AUROC	Log loss	Brier
Development	States + commands	0.8523	0.3771	0.1170
	Commands only	0.6419	0.5244	0.1711
	Training prevalence	0.5000	0.6374	0.2222
Fresh, pooled	States + commands	0.7737	0.6645	0.1865
	Commands only	0.5524	1.1081	0.2853
	Training prevalence	0.5000	0.6584	0.2327

Collection audit. Failures comprise 590 failed acquisitions, two missing-contact cases and four drops. At least one target clips in 508/512 attempts at 0.15 rad and every attempt at 0.6 rad. All eight apple zero-bias runs fail, although disturbed runs sometimes succeed. Twenty-one attempts peak within 0.5 mm of the lift cutoff, showing label sensitivity. These are task labels, not independently verified slip mechanisms.

Prediction model and splits. The predictor uses the last twenty closing samples, ending at $t = 2.96$ s before lift begins at 3.00 s. This is a 40 ms pre-lift margin, not a measured failure-warning horizon. Each sample contains 24 positions, 24 velocities and 24 command targets. Last value, mean, standard deviation and endpoint change summarize each channel, giving 288 features; commands alone give 96. No object ground truth, outcomes, event labels, future samples, injected noise, seed or identity enters the feature matrix. Predictions are computed offline, with no intervention during collection.

Training-only standardization removes near-constant features. L2 logistic regression uses L-BFGS-B; validation log loss selects $\lambda = 0.1$ from $\{0.001, 0.01, 0.1, 1\}$ for both models. Both converge. Alerts use $p_{\text{failure}} = 1 - p_{\text{success}} \geq 0.5$; log loss clips probabilities to $[10^{-12}, 1 - 10^{-12}]$. Training uses 520 attempts from two observation groups; validation and test each use 260 from one group, respectively cube and apple. All variants sharing an observation stay together. Both geometries occur in training, and the split was chosen after outcome inspection; it is an exploratory held-observation test, not an untouched unseen-object benchmark.

4 Results: ranking transfers unevenly

Measured hand response adds ranking information beyond commands on the development test (Table 1). At the fixed threshold, the full model flags 187/202 failures and 22/58 successes, leaving fifteen missed failures and a 37.9% success false-alarm rate. This is not a deployment calibration claim, and four task-scored drops cannot establish slip-warning performance.

Frozen-model fresh evaluation. A separately declared batch freezes models, normalization, inputs, threshold and scorer before collecting 2,080 attempts: 659 successes and 1,421 failures, without errors/exclusions. Familiar cube/apple and new milk carton/mug each supply two observations, four proposals per observation and 65 attempts per proposal. On the 1,040 familiar-geometry attempts, full-model AUROC is 0.9173; it flags 553/626 failures and 88/414 successes. On the 1,040 new-geometry attempts, pooled AUROC falls to 0.5588, with 582/795 failures and 185/245 successes flagged. The milk carton alone has AUROC 0.8121, but the threshold falsely flags 185 of its 245 successes. Thus pooled near-chance ranking does not describe every new object equally.

All 520 mug attempts fail from approach table collision, a category absent from training. The model flags 315 and misses 205; mug AUROC is undefined because there are no successes. Crucially, the collision precedes the 2.96 s cutoff. Its final label is valid, but its prediction is not advance warning of a future failure. This is an execution-distribution and target-timing mismatch, not simply poor

Table 2: Retrospective geometry holdouts. Newly fitted full and commands-only models; false warnings/successes and misses/failures use the full model at threshold 0.5. Mug ranking and false-warning rate are undefined.

Held geometry	Full AUROC	Cmd AUROC	False warnings	Misses
Apple	0.8238	0.6565	30/210	351/830
Milk carton	0.8345	0.6584	86/245	36/275
Cube	0.8729	0.7643	471/648	15/392
Mug	—	—	—	230/520

slip generalization. Pooling objects also mixes geometry, scene, mass and failure mode. Overall ranking improves over commands alone, yet full-model log loss is worse than constant prevalence (Table 1); ranking and probability quality should be reported separately.

Retrospective object-held-out refitting. A separately frozen analysis reuses only the 3,120 historical v2 attempts: four geometries, twelve observations and 48 proposal families. Each entire geometry is held out; among remaining geometries, the lexicographically last observation content ID validates and the others train. The same model classes, training-only preprocessing, regularization grid and fixed alert rule are refitted per fold. Corrected-controller data are excluded. Membership is label-independent, but all outcomes were previously inspected: this is retrospective development, not prospective validation or an improved score from the original frozen model. Cube ranking remains strong while 72.7% of its successes trigger warnings (Table 2).

Separate corrected-controller diagnostics. The v3 controller adjusts wrist configuration while preserving intended palm/finger geometry, with a per-step clearance guard. Of 27 checks, 26 complete; one mug stops before the full input prefix, retaining null label/score. The unchanged model warns on all 20 successful and three failed completed mugs. These reuse eight proposals/two development observations. A separate zero-bias v4 comparison warns on all 13 successes and three failures across eight wrist pairs; one label crosses the lift cutoff by less than 0.25 mm. Both wrist settings remain outside training ranges. These separate diagnostics establish repeated false alarms, not a remedy or isolated wrist causation.

5 Limitations and next steps

The frozen model sees only two training geometries; all analyses span four total geometries, twelve observations and correlated proposal/noise families. Proposal-family resampling cannot quantify variation across unseen geometries. Friction is nominal, mass is fixed within each object, and geometry/controller effects are not isolated. Simulation contacts are not validated tactile measurements. No physical transfer, dynamic mapping equivalence, useful warning lead time, recovery or multi-generator comparison is established. Compact datasets and models are preserved; raw rollouts/licensed assets are not fully redistributed.

Next, qualify one named hand/mapping/executor pipeline before broader collection, keeping mapping rejection, runtime stop and completed failure distinct. The native-hand alternative needs surface, inertia, actuation and dynamic validation. Expand HUG-only coverage across objects, grasp/approach poses, table heights, mass, friction and sensor noise, with distinct development/validation groups and whole-object holdouts. A future-event forecast requires acquisition, separately defined events, preceding causal windows, validation-selected false-alarm budgets and measured latency. New training improvements require an independently frozen evaluation; current results remain task-outcome diagnostics.

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